

Tianyi Wang

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EDUCATION

Peking University

B.S. in Biological Science

Second Major: Economics, National School of Development (NSD)

Overall GPA: 3.8/4.0

TOEFL iBT: 108/120 (R 29, L 28, S 25, W 26)

Beijing, China

09/2023 – 07/2027

PUBLICATIONS

Peer-Reviewed Workshop Papers

- **Tianyi Wang.** Knowing When to Stop: Pertura for Graph-Enforced PI Gating in Perturb-seq Agents. *ICML 2026 Workshop on Generative and Agentic AI for Biology (GenBio)*, Seoul, Korea. **Spotlight.**

Under Review

- **Tianyi Wang.** Knowing When to Claim: Pertura for Execution-Grounded Claim Promotion in Perturb-seq Agents. *Submitted to AAAI 2027.*
- Qianhui Xu, Qixiang Gui, **Tianyi Wang**, Xu Zhao, Jin Wang, Jianghui Ying, Hao Hu, Xinyi Tang, Rui Lin, Xiaoyu Yu, Xiaohui Zhou. Macrophage Panx1 Restrains Aspartate Associated Thermogenic Remodeling of White Adipose Tissue in Obesity. *Manuscript under review*, 2026.

RESEARCH EXPERIENCE

Newman Lab, Stanford University

Visiting Research Student — Advisor: Aaron M. Newman

Stanford, CA

06/2026 – Present

- Contributed to the development and evaluation of a de-novo analysisng method for spatial transcriptomic data
- Designed metrics and evaluation pipeline for fair comparison on model's performance, the ability to recover biology in spatial data and the scalability analysis: runtime, memory, scaling performance
- Benchmarked against baseline and 3 other published methods on 3 scenarios across large scale data and self-designed synthetic data
- Selected and curated the datasets used to demonstrate the method's biological interpretation and cross model validation

Zemin Zhang Lab, Peking University

Undergraduate Researcher — Advisors: Zemin Zhang, Dongfang Wang

Beijing, China

Cross-Tissue Single-Cell and Spatial Analysis of Sex and Hormone effects in MASH

12/2025 – Present

- Integrated and annotated ~2.5M single cells spanning 21 tissues in a 2×2×2 design crossing MASH, sex and hormone status, constructing a cross-tissue reference for a 5M single cell mouse atlas
- Resolved myeloid, B cell, and neutrophil populations to subtype level; identified myeloid subpopulations and gene programs whose responses to diet and sex differ across tissues
- Designed a unified compositional and pseudobulk differential expression framework used across the study, applied across all 21 tissues to separate diet, sex, and hormone effects
- Led the spatial analysis of the study across 65 Xenium sections and 8 tissues, annotated major compartments and related gene program scores to neighborhood composition and expression, resolving co-localization trends that discrete cell-type labels could not capture, validated and extended the analysis in single cell modality

Xenium 5k Spatial Atlas of the Hepatocellular Carcinoma Microenvironment

07/2025 – 11/2025

- Constructed a spatial atlas of 650,931 cells across 57 Xenium sections from an HCC cohort on combination targeted-immunotherapy; tested multiple analyzing methods and annotated 20 cell types from a ResolVI
- Compositional difference analysis and cell abundance analysis was done within each classified anatomical region, which showed distinct immune infiltration and parenchymal landscape.
- Correlated cell type fractions and cellular spatial patterns with clinical outcomes by performing survival analysis(Cox regression) on subtype proportions, identifying potential biomarkers for targeted/immunotherapy combination outcome

Pertura / PerturaBench
Independent Research — Sole Author

Beijing, China
04/2026 – 08/2026

- Built an execution-grounded authority layer for LLM agents in Perturb-seq analysis, gating every scientific claim on verified execution, analysis scope, and upstream dependencies rather than on the agent's own account of what it did
- Designed PerturaBench: 20 dependency-aware tasks over three Perturb-seq datasets and a donor-aware supplement, scoring 120 turns across three agent conditions and two prespecified repeats
- Prespecified the primary analysis and froze evaluators and thresholds before unblinding; reported clustered bootstrap confidence intervals and a sensitivity cohort, including the contrast that did not reproduce
- Reached 83.3% primary scientific fidelity against 44.4% and 38.9% for baselines, and 82.5% claim-authority success with no loss of informativeness

COURSE PROJECTS

Single-Cell Tumor Heterogeneity and Microenvironment Landscape of Hepatocellular Carcinoma

05/2025 – 06/2025

- Led a three-person team reproducing and extending a published HCC heterogeneity study; integrated 51 samples across three public scRNA-seq cohorts, recovering the reported metabolic, proliferative, and EMT malignant subtypes under independent quality control.
- trajectory and transcription factor analyses to reconstruct subtype evolution, stratified patients by subtype composition, and annotated the T/NK, fibroblast, and endothelial compartments

Spatial Domain Analysis of Pulmonary Fibrosis with Graph Neural Networks

04/2025 – 06/2025

- Built spatial graphs from 10x Xenium data and trained a GraphSAGE model to learn transcript-neighborhood embeddings; clustered the embeddings to recover spatial niches and validated them against cell-type composition.

TECHNICAL SKILLS

Languages: Python, R, Linux, Bash, ...
Single-cell & Spatial: Scanpy, squidpy, AnnData, scvi-tools, cNMF, Seurat, edgeR, ...
ML & Infrastructure: PyTorch, PyTorch Geometric, scikit-learn, SLURM on HPC, Git, ...
Platforms: 10x Chromium, 10x Xenium, MERSCOPE, Perturb-seq / CITE-seq, ...

HONORS & AWARDS

National Scholarship for Undergraduate Students (top 4%)	2024
Peking University Third-Class Scholarship (top 10%)	2025
Merit Student, Peking University (top 10%)	2024
Outstanding Student Cadre, Peking University (top 3%)	2025

LEADERSHIP & SERVICE

Student Union, School of Life Sciences, Peking University
Head of External Relations

09/2024 – 09/2025

- Led external partnerships and coordinated school-wide events including the Life Sciences New Year's Gala and Cultural Festival.

Peking University Bodybuilding Team
Member

09/2025 – Present

- Competed for Peking University at the 2026 Capital University Fitness and Bodybuilding Championships.